

Deepak Binkam

AI / Data Analytics Engineer | ML & Retrieval Systems

M.S. Computer Science graduate focused on ML, data analytics, and retrieval-based systems. Built end-to-end Python pipelines emphasizing deterministic, reproducible retrieval and quantitative evaluation (gold sets; Hit@K, MAP@5, NDCG@5, MRR). Experience with embeddings, vector databases, and REST services.

SKILLS

- **ML / Retrieval:** RAG, SentenceTransformers, ChromaDB, embeddings, Retrieval evaluation (Hit@K / MAP / MRR)
- **Programming & Data:** Python (Pandas, NumPy), SQL (MySQL; Oracle – basic), Java, C++ (JUCE)
- **Web & Backend:** REST APIs, Node.js, Express, React, Postman
- **Tools:** Streamlit, MongoDB, Power BI

PROJECTS

MSK Clinical Agentic Retrieval-Augmented QA System | Dec 2025 https://github.com/Deepak-DCB/MSK_RAG_Chatbot

- Built an agentic RAG pipeline over **20** MSKNeurology clinical essays, indexing **1,313** sentence-aware chunks in a **persistent ChromaDB** vector store.
- Implemented HTML mirroring + cleaning and **token-aware chunking** that preserves section structure and biomechanical causality; stored chunk metadata (title/section/path/token length) for traceable retrieval.
- Added an LLM controller for **query classification + biomechanics-aligned rewriting**, plus domain heuristics that promote mechanism-dense sections and demote narrative case stories.
- Built deterministic **context packing** under a fixed token budget and a Streamlit UI exposing ranked chunks, similarity scores, confidence, and latency telemetry.
- Evaluated retrieval on a **50-question gold set** using **Hit@K, MAP@5, NDCG@5, and MRR**; compared baseline retrieval vs LLM reranking and found reranking **reduced** performance, so kept the baseline as default.

Advanced Data Mining Projects | COSC 757, Towson University (Spring 2025)

- **Storm Severity and Cost Prediction (NOAA):** Built an end-to-end pipeline (cleaning, features, modeling) and compared XGBoost vs Random Forest baselines; reported performance using appropriate metrics (RMSE/MAE for cost; F1/AUC for severity classes).
- **Hybrid Nanofluid Density Prediction (Kaggle):** Developed regression workflow with dimensionality reduction and clustering for structure discovery, and evaluated with regression error metrics and cross-validation.
- **Bitcoin Movement Classification:** Engineered time-based features (lags/rolling windows) and benchmarked logistic regression, decision trees, and KNN; reported classification metrics (accuracy/F1/AUC) on held-out data.

Towson University Olympics Website | Software Development Lifecycle

- Engaged in a full-cycle software development project, using an agile, sprint-based workflow.
- Contributed to requirements analysis, use case development, and architectural modeling to align system design with user needs.
- Translated functional requirements into practical system components while supporting iterative development and design tradeoffs.

EDUCATION

Towson University, Towson, MD

Master of Science in Computer Science | 2025 | 3.87 GPA

Relevant Courses: Data Structures, Data Mining, Software Testing/Maintenance, Advanced Web Development

University of Maryland, College Park

Bachelor of Science in Information Science | 2022

Relevant Courses: Object-Oriented Programming, User-Centered Design, Human-Centered Cybersecurity
